

STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

A Mini Project Report Submitted for Machine Learning Lab

Submitted By

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Abstract

Machine Learning is one of the most important technologies in modern computer science. It allows systems to learn patterns from data and make intelligent predictions. In the education sector, Machine Learning can help institutions analyze student performance and predict academic outcomes.

This project focuses on predicting student final marks using Machine Learning techniques. The prediction is based on several academic parameters such as study hours, attendance percentage, assignment scores, and internal marks. The project uses the Linear Regression algorithm to train the model and predict student performance.

The main objective of this project is to demonstrate the practical implementation of Machine Learning concepts in the education domain. The system helps teachers and institutions identify weak students and take corrective measures to improve academic performance.

This project also explains important Machine Learning processes such as data preprocessing, train-test splitting, model training, prediction, and evaluation.

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from historical data and make predictions without explicit programming. It is widely used in many domains such as healthcare, banking, education, transportation, and e-commerce.

In the education sector, Machine Learning can help improve learning systems by analyzing student data and predicting academic performance. Educational institutions collect a large amount of data every semester, including attendance, assignment scores, internal marks, and examination results. Analyzing this data manually is difficult and time-consuming.

Student performance prediction is one of the important applications of Machine Learning in education. It helps institutions identify students who may perform poorly in examinations. Early prediction enables teachers to provide proper guidance and support.

This project implements a Machine Learning model that predicts final student marks based on academic performance indicators. The Linear Regression algorithm is used because it is simple, efficient, and suitable for predicting numerical values.

2. Problem Statement

Educational institutions often face difficulties in monitoring the academic progress of students. Manual analysis of student performance is time-consuming and may not provide accurate predictions.

The problem addressed in this project is to develop a Machine Learning model that can predict student final marks using factors such as:

- Study Hours
- Attendance Percentage
- Assignment Scores
- Internal Examination Marks

The predicted results can help teachers identify weak students and improve their performance.

3. Objectives

The objectives of this project are:

1. To understand the concept of Machine Learning.
2. To implement Linear Regression practically.
3. To predict student performance using academic data.
4. To analyze the relationship between study habits and marks.
5. To evaluate model performance using error metrics.

4. Scope of the Project

The scope of this project includes:

- Educational data analysis
- Student performance prediction
- Academic performance monitoring
- Learning Machine Learning algorithms practically

The project can be extended in the future by:

- Using large real-world datasets
- Implementing advanced algorithms
- Developing a web-based application

Adding graphical dashboards

5. Literature Review

Several researchers have worked on educational data mining and student performance analysis. Machine Learning techniques are widely used to predict student outcomes.

Some commonly used algorithms are:

- Linear Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Neural Networks

Among these algorithms, Linear Regression is simple and effective for predicting continuous values such as marks and percentages.

Research studies show that attendance, study hours, and assignment completion significantly affect student academic performance.

6. System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 MB free space

Software Requirements

- Operating System: Windows/Linux
- Python 3.x
- Jupyter Notebook or VS Code

Python Libraries Used

- pandas
- numpy
- matplotlib
- scikit-learn

7. Methodology

The project follows the standard Machine Learning workflow.

Step 1: Data Collection

A dataset containing student academic information is created. The dataset includes:

- Study Hours
- Attendance
- Assignment Scores
- Internal Marks
- Final Marks

Step 2: Data Preprocessing

Data preprocessing is performed to prepare the data for training.

The following operations are carried out:

- Handling missing values
- Separating input and output variables
- Organizing data into tabular format

Step 3: Train-Test Split

The dataset is divided into:

- Training Data (80%)
- Testing Data (20%)

Training data is used to train the model, while testing data is used to evaluate the performance.

Step 4: Model Training

The Linear Regression algorithm is trained using the training dataset.

The algorithm learns the relationship between input features and target values.

Step 5: Prediction

The trained model predicts student marks for test data.

Step 6: Evaluation

The performance of the model is evaluated using Mean Absolute Error (MAE).

Lower error values indicate better prediction accuracy.

8. Machine Learning Algorithm

Linear Regression

Linear Regression is a supervised Machine Learning algorithm used for predicting continuous numerical values.

It establishes a linear relationship between independent variables and dependent variables.

The equation of Linear Regression is:

$$Y = mX + c$$

Where:

- Y = Predicted Output
- X = Input Variable
- m = Slope
- c = Constant

Advantages of Linear Regression:

- Simple to implement
- Fast training
- Easy interpretation
- Works well for small datasets

9. Dataset Description

The dataset used in this project contains academic information of students.

Attribute	Description
StudyHours	Number of study hours
Attendance	Attendance percentage
AssignmentScore	Assignment marks
InternalMarks	Internal examination marks
FinalMarks	Final examination marks

Sample Dataset:

StudyHours	Attendance	AssignmentScore	InternalMarks	FinalMarks
2	60	50	40	42
5	75	65	55	60
8	85	78	68	76
10	92	90	80	90

10. Python Implementation

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error

# Create dataset
data = {
    'StudyHours': [2,3,4,5,6,7,8,9,10,11],
    'Attendance': [60,65,70,75,80,82,85,88,92,95],
    'AssignmentScore': [50,55,60,65,70,75,78,82,90,95],
    'InternalMarks': [40,45,50,55,60,65,68,72,80,85],
    'FinalMarks': [42,48,55,60,66,72,76,81,90,96]
}

# Create dataframe
df = pd.DataFrame(data)

# Features
X = df[['StudyHours', 'Attendance',
        'AssignmentScore', 'InternalMarks']]

# Target
y = df['FinalMarks']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Create model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)
```

```
# Predict output
predictions = model.predict(X_test)

# Print predictions
print("Predicted Marks:")
print(predictions)

# Error calculation
mae = mean_absolute_error(y_test, predictions)
print("Mean Absolute Error:", mae)

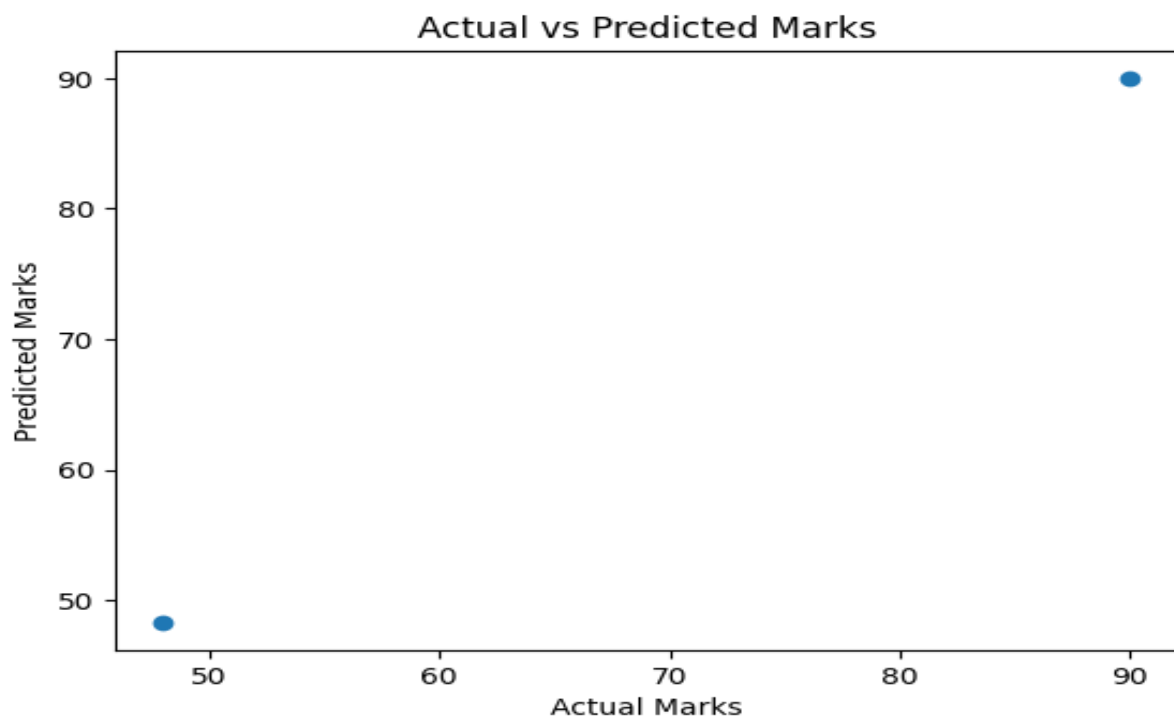
# Graph
plt.scatter(y_test, predictions)
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.title("Actual vs Predicted Marks")
plt.show()
```

11. Output

Predicted Marks:

[90.00190978 48.27637783]

Mean Absolute Error: 0.1391438084131842



12. Advantages

Advantages of the project:

1. Easy to implement
2. Helps identify weak students
3. Reduces manual analysis effort
4. Improves academic monitoring
5. Demonstrates practical Machine Learning concepts

13. Limitations

Limitations of the project:

1. Uses a small dataset
2. Prediction accuracy depends on data quality
3. Linear Regression may not capture complex relationships

Real-world datasets may contain noisy data

14. Applications

Applications of Student Performance Prediction:

- Educational institutions
- Online learning platforms
- School management systems
- Academic analytics
- Student counseling systems

15. Future Scope

The project can be improved in the following ways:

1. Use large real-world datasets
2. Implement advanced algorithms like Random Forest and XGBoost
3. Create a web application using Flask or Streamlit
4. Add data visualization dashboards
5. Integrate real-time student monitoring systems

16. Conclusion

This project successfully demonstrates the implementation of Machine Learning for predicting student performance. The Linear Regression algorithm is used to predict final marks based on study hours, attendance, assignment scores, and internal marks.

The project highlights how Machine Learning can be applied in the education domain to improve academic analysis and decision-making.

The system can help teachers identify students requiring additional support and improve overall educational outcomes.

17. References

1. Python Documentation – <https://www.python.org/>
 2. Scikit-learn Documentation – <https://scikit-learn.org/>
 3. Pandas Documentation – <https://pandas.pydata.org/>
 4. Machine Learning Concepts by Andrew Ng
- Educational Data Mining Research Papers